

Table 7: Snapshot of puzzles in the dataset for k=2

| Puzzle                                                                                                                                                                   | Question                         | Gender                                                 | Answer   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------------------|----------|
| <i>Charles's son Christopher entered rehab for the ninth time at the age of thirty. Randolph had a nephew called Christopher who had n't seen for a number of years.</i> | Randolph is the _____ of Charles | Charles: male,<br>Christopher: male,<br>Randolph: male | brother  |
| <i>Randolph and his sister Sharon went to the park. Arthur went to the baseball game with his son Randolph</i>                                                           | Sharon is the _____ of Arthur    | Arthur: male,<br>Randolph: male,<br>Sharon: female     | daughter |
| <i>Frank went to the park with his father, Brett. Frank called his brother Boyd on the phone. He wanted to go out for some beers.</i>                                    | Brett is the _____ of Boyd       | Boyd: male,<br>Frank: male,<br>Brett: male             | father   |

Table 8: Snapshot of puzzles in the dataset for k=3

| Puzzle                                                                                                                                                                                                                       | Question                       | Gender                                                               | Answer      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------|-------------|
| <i>Roger was playing baseball with his sons Sam and Leon. Sam had to take a break though because he needed to call his sister Robin.</i>                                                                                     | Leon is the _____ of Robin     | Robin: female,<br>Sam: male,<br>Roger: male,<br>Leon: male           | brother     |
| <i>Elvira and her daughter Nancy went shopping together last Monday and they bought new shoes for Elvira's kids. Pedro and his sister Allison went to the fair. Pedro's mother, Nancy, was out with friends for the day.</i> | Elvira is the _____ of Allison | Allison: female,<br>Pedro: male,<br>Nancy: female,<br>Elvira: female | grandmother |
| <i>Roger met up with his sister Nancy and her daughter Cynthia at the mall to go shopping together. Cynthia's brother Pedro was going to be the star in the new show.</i>                                                    | Pedro is the _____ of Roger    | Roger: male,<br>Nancy: female,<br>Cynthia: female,<br>Pedro: male    | nephew      |

Table 9: Snapshot of puzzles in the dataset for k=4

| Puzzle                                                                                                                                                                                                                                                                                                                                          | Question                                     | Gender                                                                                                                     | Answer   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------|
| <i>Celina</i> has been visiting her sister, <i>Fran</i> all week. <i>Fran</i> is also the daughter of <i>Bethany</i> . <i>Ronald</i> loves visiting his aunt <i>Bethany</i> over the weekends. <i>Samuel</i> 's son <i>Ronald</i> entered rehab for the ninth time at the age of thirty.                                                        | <i>Celina</i> is the _____ of <i>Samuel</i>  | <i>Samuel</i> :male,<br><i>Ronald</i> :male,<br><i>Bethany</i> :female,<br><i>Fran</i> :female,<br><i>Celina</i> :female   | niece    |
| <i>Celina</i> adores her daughter <i>Bethany</i> . <i>Bethany</i> loves her very much, too. <i>Jackie</i> called her mother <i>Bethany</i> to let her know she will be back home soon. <i>Thomas</i> was helping his daughter <i>Fran</i> with her homework at home. Afterwards, <i>Fran</i> and her sister <i>Jackie</i> played Xbox together. | <i>Celina</i> is the _____ of <i>Thomas</i>  | <i>Thomas</i> :male,<br><i>Fran</i> :female,<br><i>Jackie</i> :female,<br><i>Bethany</i> :female,<br><i>Celina</i> :female | daughter |
| <i>Raquel</i> is <i>Samuel</i> 's daughter and they go shopping at least twice a week together. <i>Kenneth</i> and her mom, <i>Theresa</i> , had a big fight. <i>Theresa</i> 's son, <i>Ronald</i> , refused to get involved. <i>Ronald</i> was having an argument with her sister, <i>Raquel</i> .                                             | <i>Samuel</i> is the _____ of <i>Kenneth</i> | <i>Kenneth</i> :male,<br><i>Theresa</i> :female,<br><i>Ronald</i> :male,<br><i>Raquel</i> :female,<br><i>Samuel</i> :male  | father   |

Table 10: Snapshot of puzzles in the dataset for k=5

| Puzzle                                                                                                                                                                                                                                                                                                                                                                         | Question                         | Gender                                                                                         | Answer  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------|---------|
| <i>Steven's son is Bradford. Bradford and his father always go fishing together on Sundays and have a great time together. Diane is taking her brother Brad out for a late dinner. Kristin, Brad's mother, is home with a cold. Diane's father Elmer, and his brother Steven, all got into the rental car to start the long cross-country roadtrip they had been planning.</i> | Bradford is the _____ of Kristin | Kristin:female,<br>Brad:male,<br>Diane:female,<br>Elmer:male,<br>Steven:male,<br>Bradford:male | nephew  |
| <i>Elmer went on a roadtrip with his youngest child, Brad. Lena and her sister Diane are going to a restaurant for lunch. Lena's brother Brad is going to meet them there with his father Elmer. Brad can't stand his unfriendly aunt Lizzie.</i>                                                                                                                              | Lizzie is the _____ of Diane     | Diane:female,<br>Lena:female,<br>Brad:male,<br>Elmer:male,<br>Lizzie:female                    | aunt    |
| <i>Ira took his niece April fishing Saturday. They caught a couple small fish. Ronald was enjoying spending time with his parents, Damion and Claudine. Damion's other son, Dennis, wanted to come visit too. Dennis often goes out for lunch with his sister, April.</i>                                                                                                      | Ira is the _____ of Claudine     | Claudine:female,<br>Ronald:male,<br>Damion:male,<br>Dennis:male,<br>April:female,<br>Ira:male  | brother |